

**Ex.No:**

**EXPLORATORY DATA ANALYSIS (EDA) – DATA INSPECTION,  
FILTERING, STATISTICAL ANALYSIS AND CORRELATION**

**Date:**

**AIM:**

To create and analyze various data visualizations such as Line Charts, Bar Charts, Pie Charts, Histograms, Scatter Plots, and Box Plots using Matplotlib for effective data interpretation.

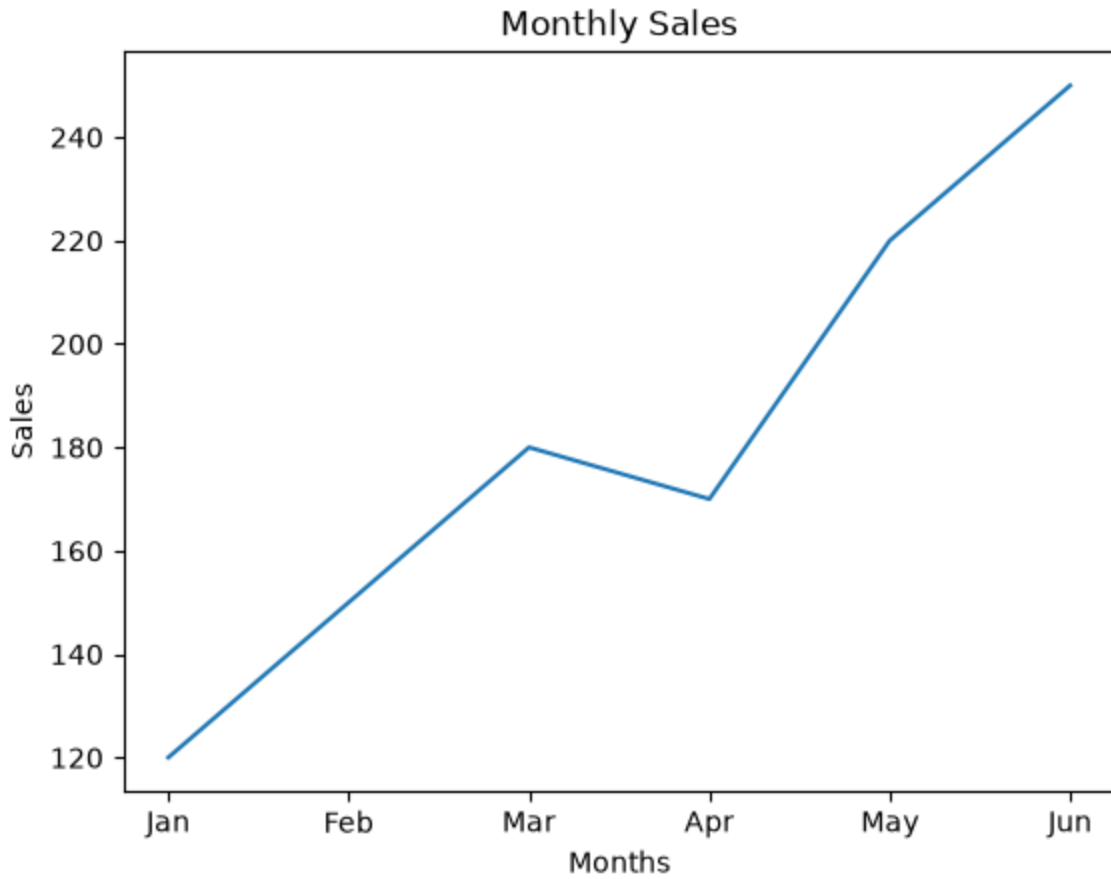
**ALGORITHM:**

**CODE:**

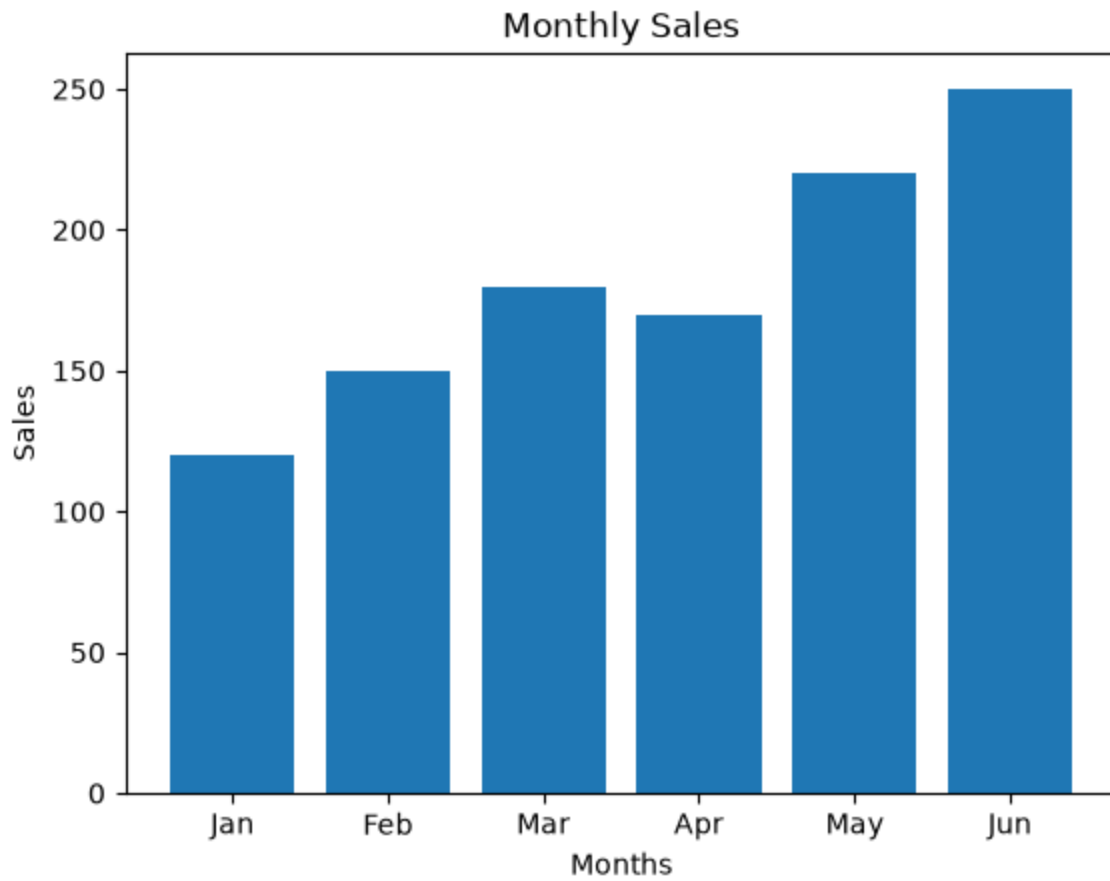
```
import matplotlib.pyplot as plt

months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
sales = [120, 150, 180, 170, 220, 250]

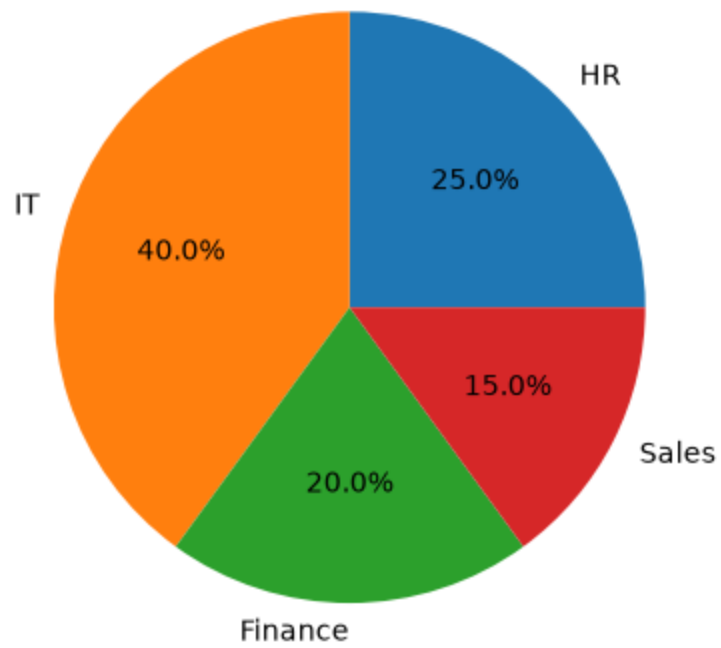
plt.plot(months, sales)
plt.title("Monthly Sales")
plt.xlabel("Months")
plt.ylabel("Sales")
plt.show()
```



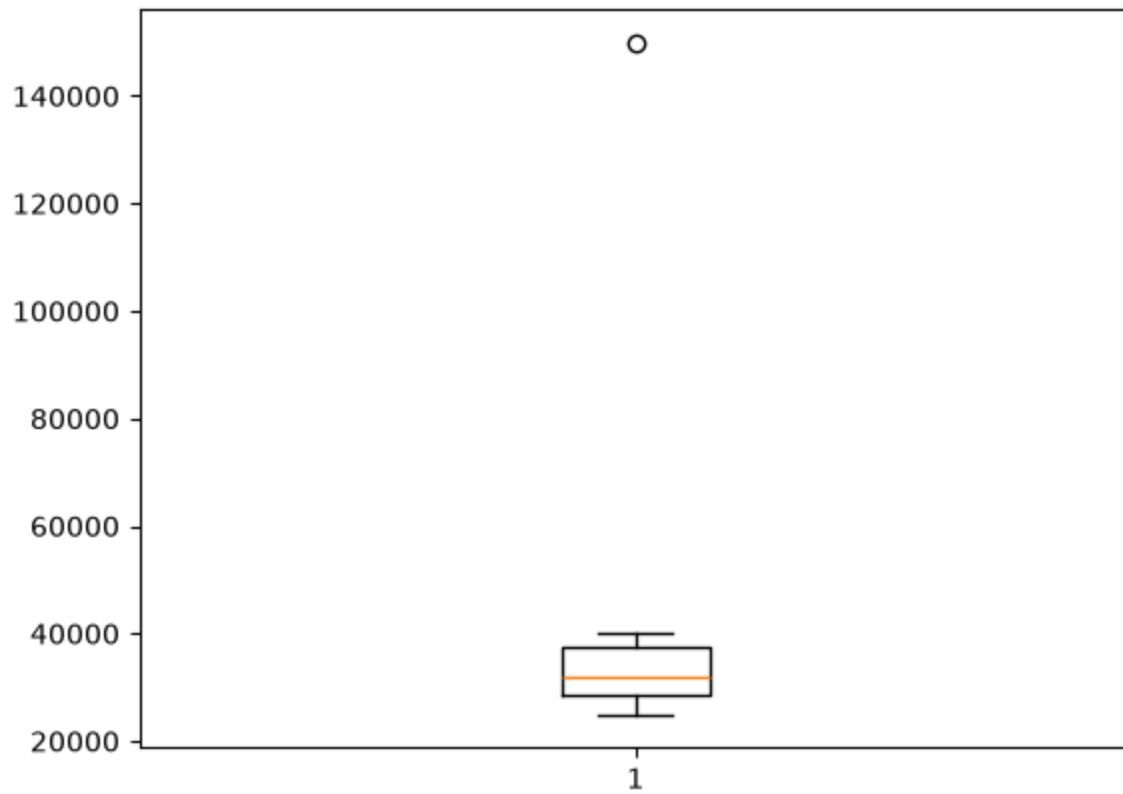
```
plt.bar(months,sales)
plt.title("Monthly Sales")
plt.xlabel("Months")
plt.ylabel("Sales")
plt.show()
```



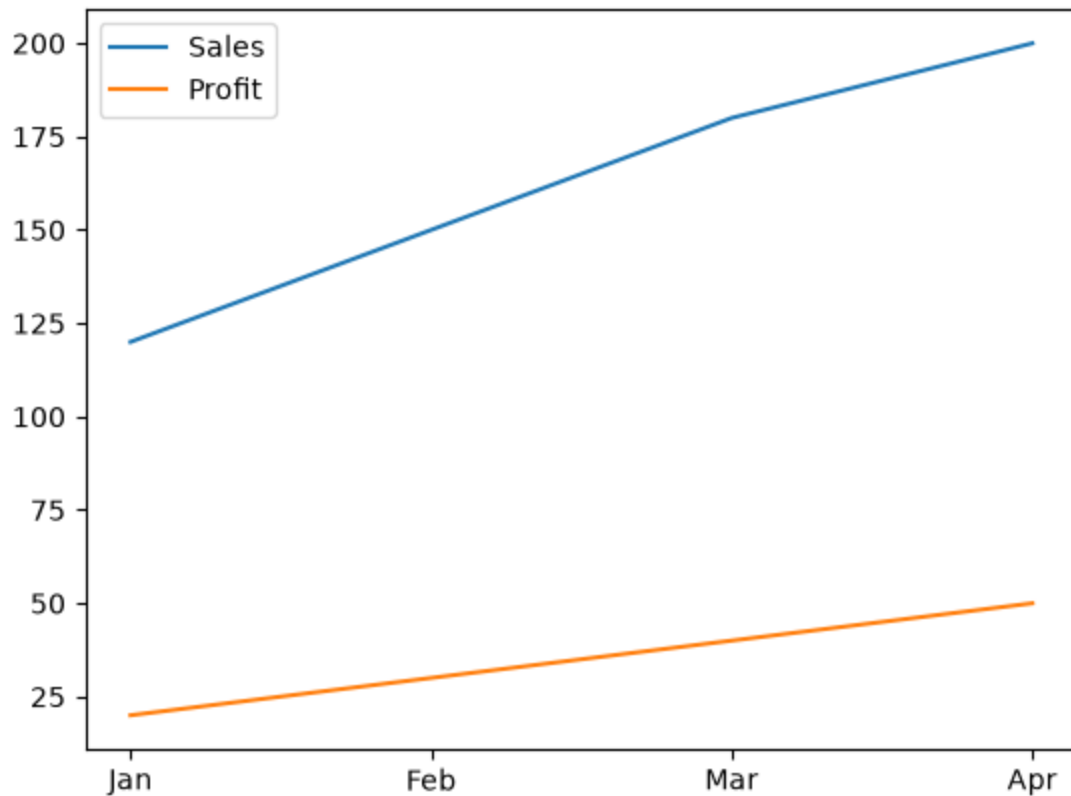
```
departments = ["HR", "IT", "Finance", "Sales"]
employees = [25, 40, 20, 15]
plt.pie(employees,
labels=departments,
autopct='%1.1f%%')
plt.show()
```



```
salary=[25000,27000,30000,  
         32000,35000,  
         40000,150000]  
plt.boxplot(salary)  
plt.show()
```



```
months=["Jan", "Feb", "Mar", "Apr"]
sales=[120,150,180,200]
profit=[20,30,40,50]
plt.plot(months,sales,label="Sales")
plt.plot(months,profit,label="Profit")
plt.legend()
plt.show()
```



## RESULT:

Thus, various visualization techniques such as Line Charts, Bar Charts, Pie Charts, Histograms, Scatter Plots, and Box Plots were successfully created using Matplotlib. The graphical representations helped in understanding trends, distributions, comparisons, and relationships within the dataset.